Circular Economy (CE) Business Model Selection Using Multi-Criteria Decision-Making (MCDM) Methods MOORA and EDAS

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Abstract

With the robust growth in artificial intelligence boom in India, it is becoming crucial for government and private sectors to shift towards sustainable practices to manage existing resources. The growth in numerous organizations and development of new policies will considerably lead to environmental degradation in forthcoming years. Thus, the primary aim of this paper is to assess different circular practices that will help organizations in India to drive their workforce towards productivity. Circular Economy Business Models focuses on enhancing sustainability via evaluating various Circular practices. This paper aims to rank these practices using Multi-Criteria Decision-Making methods: "multi-objective optimization on the basis of ratio analysis" (MOORA) & "Distance from average solution" (EDAS). In both cases we obtain the rank one for the circular practice (CE15), indicating adoption of R's (reduce, reuse, remanufacture, recycle) practices for sustainable way of operating. MOORA and EDAS have a very close relation for calculating rankings with a very high positive correlation value very close to one. Both MCDM approaches yield same output rank for "R's practices" validating and reinforcing one another. The R practice selected is essential to adopt and implement for an organization to move towards CE.

Keywords:Circular, Economy, MCDM, EDAS, MOORA

1.0 Introduction

The increasing requirement of circular models for a sustainable environment indicates the importance of CE practices in various economic contexts [1]. To evaluate the impact of the judgements given by various experts for the determination of a beneficial CE practice, researchers use multi-criteria decision-making (MCDM) methods. Integrating MCDM methods for a study in a qualitative assessment showcases the robustness of MCDM methods [2]. There is a necessity of exploring MCDM methods like MOORA (Multi-Objective Optimization on the basis of Ratio Analysis) and EDAS (Evaluation Based on Distance from Average Solution) for comparative analysis in while dealing with circular economy area [3]. Considering these policies and practices while using the techniques for empirical study is required to validate the applicability of MCDM methods [4]. Focusing on circular economy, rather than linear economy, is necessary for overall development [1]. Optimizing resources and minimizing waste for global development is the main aim for Sustainable Development Goals (SDGs) [5]. Robust technique is required while handling multiple criteria [2]. Multi-criteria decision making (MCDM) techniques like MOORA and EDAS have been extensively used for ranking in recent research [3][6]. The efficiency of MOORA MCDM method can be seen in

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Solving variety of problems in Manufacturing and industrial sectors [7]. Similarly, by judging alternatives by how far they differ from an average answer, the EDAS MCDM technique is helpful for determining and offering answers to end decision-making scenarios [8].

1.1 Aims and Objectives

With the increase in population of India, challenges related to resource depletion and environmental degradation have inclined the nation to shift from traditional linear economic model towards closed-loop circular economy system [19]. Certain initiatives have been undertaken that aim to minimize environmental impact, by optimizing production and consumption, and supporting India's Sustainable Development Goals (SDGs). Multi-Criteria Decision-Making (MCDM) techniques are beneficial in evaluating various circular economy (CE) practices. Defining policy priorities involves a number of considerations. MCDM techniques expedites the decision-making process by assisting policymakers examine several options and the implications in order to choose the best circular economy tactics [18]. This research uses EDAS (Evaluation based on Distance from Average Solution) and MOORA (Multi-Objective Optimization on the basis of Ratio Analysis) to rate several policies. Through the work in this research, MOORA (Multi-Objective Optimization on the basis of Ratio Analysis) and EDAS (Evaluation based on Distance from Average Solution) are applied to rank multiple policies. The study concludes ranking CE policies, promoting sustainable development.

Using these two techniques, MOORA and EDAS this paper access circular economic business models for enhancing sustainability. This paper addresses ranking Circular Economy (CE) practices using multi-criteria decision-making (MCDM) techniques. Taking a structured decision-based approach, two robust MCDM methods—MOORA (Multi-Objective Optimization on the basis of Ratio Analysis) and EDAS (Evaluation Based on Distance from Average Solution)—are used to obtain rankings of 15 CE alternatives. The dataset comprises expert evaluations, and the selection and ranking of CE alternatives is done on a 5-point scale, treated as crisp input values. Section 2 introduces to the dataset used, where expert judgments are used as performance scores across multiple criteria. Section 3 explains the concepts of MCDM and details the mathematical formulations of the MOORA and EDAS methods with flowcharts and pseudocodes. In Section 4, rankings derived from the MCDM techniques are shown, and sensitivity analysis is used to compare the outcomes. Conclusions and consequences for these 15 policies for sustainable practice are given in Section 5.

2. Literature Review

For overcoming challenges in sustainable development, including consumption and production, circular economy (CE) is regarded as a pivotal factor for driving the required changes [1]. There is a need for organized framework to work on unstructured data for implementing circular economy (CE) practices [11]. For improving performance, Frameworks such as ISO 59020 and BS 8001:2017 were proposed for assessing circularity performance [12]. The MOORA (Multi-Objective Optimization on the basis of Ratio Analysis) technique initially was proposed by Brauers and Zavadskas, and became popular due to its robustness to handle both benefit and cost criteria [3]. MOORA is useful for complex decision-making comparisons that are applied in manufacturing domains [7]. The reliability of MOORA for expert judgements in diverse criteria is proven for ranking in sustainability evaluations [3]. Alongside hybrid methods like COBRA, MOORA has shown strong performance in waste management [14]. EDAS (Evaluation Based on Distance from Average Solution) uses the deviation via average performance for ranking alternatives during decision-making process [6]. The productive solutions of EDAS in decision making while solving environmental problems is evident, particularly in sustainability domains [10]. Practical data is necessary for CE strategies, involving supply chains or waste handling to indicate assessment of CE models [15]. Reliability of the two MCDM methods can be verified by simultaneously integrating these techniques in evaluations [14][15].

3. Circular Economy (CE) practices

Dataset used in this research comprises of 15 circular practices and the judgement given by 5 experts.

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Table 1: Decision Matrix [4]

CE Items	Expert 1	Expert 2	Expert 3	Expert 4	Expert 5
CE_1	4	3	4	5	4
CE_2	2	3	4	3	4
CE_3	2	3	2	2	3
CE_4	3	2	2	2	2
CE_5	3	4	3	3	4
CE_6	3	4	3	3	2
CE_7	3	3	2	2	3
CE_8	1	2	1	2	1
CE_9	2	3	3	2	3
CE_{10}	4	4	4	5	5
CE_{11}	5	4	5	5	4
CE_{12}	4	5	4	4	4
CE_{13}	2	2	1	2	3
CE_{14}	4	4	5	5	5
CE_{15}	3	4	5	5	3

Weights given to each expert are as follows: [0.2, 0.2, 0.2, 0.2, 0.2]

Table 2: CE values Description [1]

CE Code	**Circular Practice**	**Brief Description**
CE_1	Designing products for circularity	The product should be designed to reduce material and energy consumption and enable reuse/recovery at the component level.
CE_2	Environmental criteria for supplier selection	Supplier selection considers not only economic but also environmental criteria.
CE_3	Processes design for waste minimisation	The production process is designed and adopted to create minimal waste throughout the value chain.
CE_4	Renewable materials and energy utilisation	Renewable materials and energy are used in supply chain operations, especially during production.
CE_5	Workforce training related to circular practices	Training is provided to the workforce on circular practices and environmental issues.
CE_6	Reuse of energy and water	Energy and water are reused across the supply chain to support circular economy goals.
CE_7	Implementation of reverse logistics	Reverse logistics networks are established to collect products from users after usage and end-of-life.
CE_8	Cascading use of components and materials	Enhancing the cascading use of components as well as materials.
CE_9	Performance evaluation considering environmental factors	Internal evaluation systems include environmental factors for better performance.

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CE_{10}	Develop the circular culture	Organizations are encouraged to develop a circular culture through circular management and monitoring of CE activities.
CE_{11}	Consumer awareness	Create awareness among customers regarding circular practices and their environmental impact.
CE_{12}	Awareness among the supply chain partners	Raise awareness among supply chain partners to adopt CE practices for effective implementation and sustainable development.
CE_{13}	Cross-functional collaboration	Foster cross-functional collaboration and cooperation for environmental improvements.
CE_{14}	Legislation and policies	Governments should pass legislation and policies to encourage and regulate the adoption of circular practices.
CE_{15}	R's practices	Organizations should adopt the R's (reduce, reuse, remanufacture, recycle) to transition toward circular economy principles.

Table 2 presents set of 15 Circular Economy (CE) practices that are ranked by different experts based on various criteria for enhancing productive close loop circular economy for sustainable development [16]. This table and the circular economy practices showcase transition from a traditional to a closed loop model, emphasizing reusing, repairing, and recycling for an enhanced system [16]. Involving these circular practices in a business operation has proven to be in producing results for a sustainability environment, while driving innovation and efficiency across supply chains [17]. Shahbaz Khan and Haleem [4] uses CE practices for the new economies in circular systems. In previous research studies, frameworks have been used including methodologies for evaluating CE practices [18]. Ranking CE practices with robust MCDM techniques can be useful for an optimized approach that can facilitate a sustainable development.

3.3.1 Multi-Objective Optimization on the basis of Ratio Analysis

A decision matrix comprising ratings of 15 Circular Economy (CE) options (CE_1 to CE_{15}) by five expert evaluators (Expert 1 to Expert 5) is created in order for establishing the rankings of CE alternatives using the MOORA (Multi-Objective Optimization on the basis of Ratio Analysis) strategy. Vector normalization, which divides each item by the square root of the sum of squares of all entries in that column, is applied to normalize each criterion (expert rating column). Normalized values are assigned equal weights. The weighted normalized values for each of the criteria are added up in order to calculate the MOORA score for each CE alternative [19]. Following that, the options are arranged according to their MOORA scores, where higher scores denote more advantageous CE practices.

(MOORA) Variables involved

Variable Definition

D: Decision matrix with alternatives (rows) and criteria (columns)

x_ij: Performance value of alternative i under criterion j

x_ij_normalized: Normalized value of x_ij using vector normalization

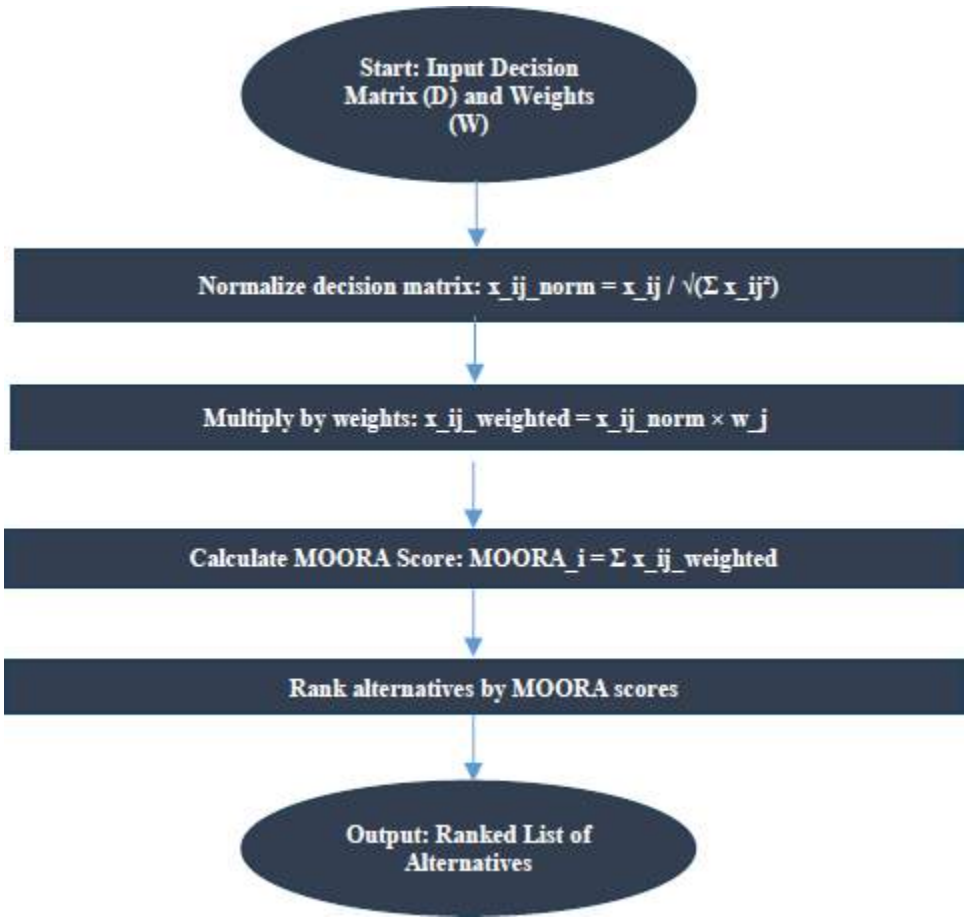
w_j: Weight assigned to the j-th criterion

x_ij_weighted: Weighted normalized value: x_ij_normalized × w_j
MOORA_score_i: Final score of the i-th alternative calculated
Σ (Sigma): Denotes summation over specified elements (e.g., all criteria j)

Table 3: MOORA Rankings [2]

PSEUDOCODE:
1. Input: Decision matrix D (alternatives × criteria), and criteria weights W.
2. Normalize the decision matrix using vector normalization:
   For each element x_ij in D:
   x_ij_normalized = x_ij / sqrt (sum (x_ij^2 for all i))
3. Multiply normalized matrix by corresponding weights:
   x_ij_weighted = x_ij_normalized × w_j
4. Calculate MOORA score for each alternative:
   MOORA_score_i = sum (x_ij_weighted for all j)
5. Rank alternatives in descending order of MOORA scores.
6. Output: Ranked list of alternatives based on MOORA scores.

Flow chart 1: (MOORA) Program algorithm [9]



3.3.2 Evaluation Based on Distance from Average Solution

The CE alternatives are ranked using the decision matrix and the EDAS (Evaluation Based on Distance from Average Solution) method. The Average Solution (AS) is the mean value for every factor (expert) across all CE choices. The Positive Distance from Average (PDA) and the Negative Distance from Average (NDA) are computed for every value in the decision matrix. The weighted positive and negative distances are derived by multiplying the PDA and NDA values by their associated weights. The weighted PDA (SP) and weighted NDA (SN) sums are obtained for each option. Higher appraisal scores indicate greater performance compared to the average for all CE practices [22]. The alternatives are arranged in descending order of assessment scores.

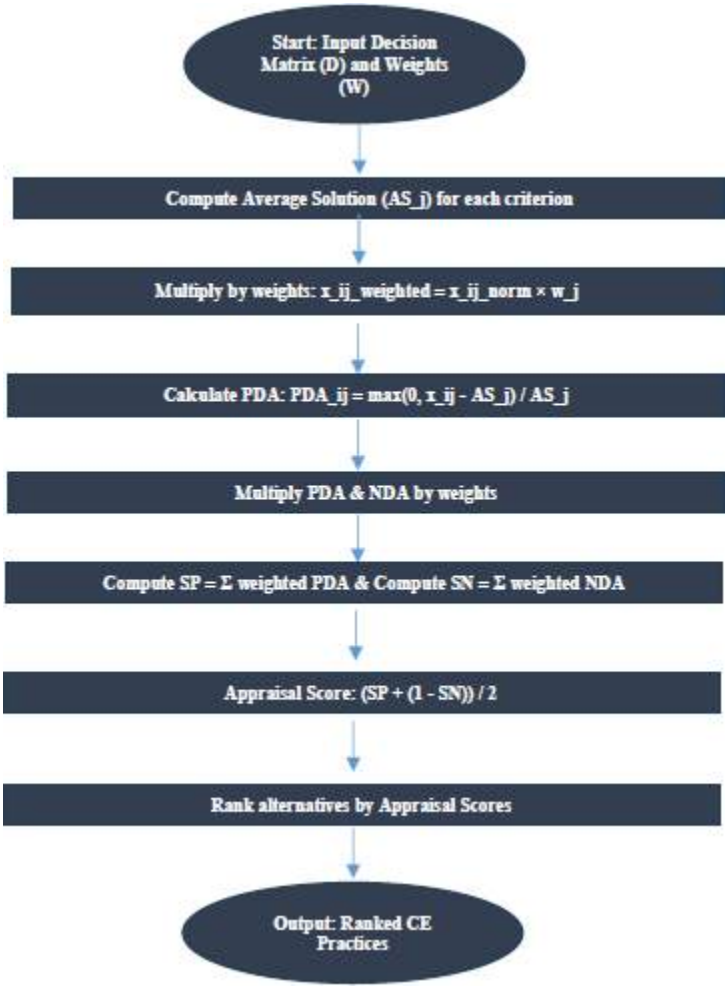
(EDAS) Variables involved
D: Decision matrix with alternatives (rows) and criteria (columns)
x_ij: Performance value of alternative i under criterion j
AS_j: Average value of the j-th criterion across all alternatives
PDA_ij: Positive Distance from Average for x_ij: max (0, x_ij - AS_j) / AS_j
NDA_ij: Negative Distance from Average for x_ij: max (0, AS_j - x_ij) / AS_j
w_j: Weight assigned to the j-th criterion
SP_i: Sum of weighted PDA values for the i-th alternative
SN_i: Sum of weighted NDA values for the i-th alternative
Appraisal_i: Final EDAS score: (SP_i + (1 - SN_i)) / 2
Σ (Sigma): Denotes summation over specified elements (e.g., all criteria j)

Table 4: EDAS Rankings [5]

PSEUDOCODE:
1. Input: Decision matrix D (alternatives × criteria), and criteria weights W.
2. Compute the Average Solution (AS) for each criterion.
3. Calculate Positive Distance from Average (PDA):
   PDA_ij = max(0, (x_ij - AS_j)) / AS_j
4. Calculate Negative Distance from Average (NDA):
   NDA_ij = max(0, (AS_j - x_ij)) / AS_j
5. Compute weighted PDA and NDA:
   weighted_PDA_ij = PDA_ij × w_j
   weighted_NDA_ij = NDA_ij × w_j
6. Calculate total SP and SN for each alternative:
   SP_i = sum(weighted_PDA_ij for all j)
   SN_i = sum(weighted_NDA_ij for all j)
7. Compute appraisal score: Appraisal_i = (SP_i + (1 - SN_i)) / 2
8. Rank alternatives in descending order of appraisal scores.
9. Output: Ranked list of alternatives based on EDAS scores.

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Flow chart 2: (EDAS) Program algorithm [10]



4.0 Ranking CE Practices

The decision matrix, formed after five experts evaluated fifteen Circular Economy (CE) practices, was analysed using Multi-Criteria Decision-Making (MCDM) techniques: MOORA and EDAS. Both methods consistently ranked R's practices (CE_{15}). The results shows that the "R's practice", is the most impactful alternative. Rankings calculated using MOORA and EDAS imply that both the methods have similar correlation, and are reliable approaches. In the MOORA method, R's practices (CE_{15}) achieved the highest score with normalized score of 0.336554. R's practices (CE_{15}) had the highest appraisal score of 0.871190, and the EDAS approach calculated the Positive Distance from Average (SP) and Negative Distance from Average (SN) as 0.759401 and 0.017021, respectively.

Table 5: Ranking MOORA vs EDAS

MOORA			EDAS				
CE Item	MOORA Score	Rank	CE Item	SP	SN	Appraisal Score	Rank
CE15	0.336554	1	CE15	0.759401	0.017021	0.871190	1
CE1	0.334047	2	CE1	0.726610	0.008696	0.858957	2
CE12	0.317930	3	CE12	0.726610	0.139130	0.793740	3
CE14	0.317214	4	CE14	0.597628	0.017021	0.790344	4
CE13	0.283408	5	CE13	0.452172	0.144294	0.653940	5
CE7	0.250427	6	CE7	0.174026	0.140003	0.517012	6
CE2	0.233544	7	CE2	0.167781	0.266403	0.450689	7
CE3	0.231745	8	CE3	0.119729	0.237627	0.441231	8
CE10	0.217407	9	CE10	0.037662	0.270437	0.383612	9
CE9	0.217123	10	CE9	0.145455	0.381948	0.381753	10
CE5	0.199079	11	CE5	0.000000	0.381561	0.309219	11
CE4	0.199079	11	CE4	0.000000	0.381561	0.309219	11
CE6	0.184104	13	CE6	0.090091	0.512583	0.248543	13
CE8	0.183317	14	CE8	0.000000	0.509221	0.245390	14
CE11	0.183317	14	CE11	0.000000	0.509221	0.245390	14

5.0 Sensitivity analysis

To confirm the consistency in outcome, weights are altered in the following manner. Following the algorithm designed initially the rankings were again calculated with the same decision matrix for the 15 CE practices.

CE Items	Expert 1	Expert 2	Expert 3	Expert 4	Expert 5
CASE 1	0.2	0.2	0.2	0.2	0.15
CASE 2	0.2	0.2	0.2	0.15	0.2
CASE 3	0.2	0.2	0.15	0.2	0.2
CASE 4	0.2	0.15	0.2	0.2	0.2
CASE 5	0.15	0.2	0.2	0.2	0.2

The rankings obtained are as follows:

CE Items	Case 1	Case 2	Case 3	Case 4	Case 5
CE1	1	2	1	1	2
CE2	8	7	7	7	7
CE3	7	8	8	8	8
CE4	11	11	11	11	11
CE5	11	11	11	11	11

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CE6	13	13	13	13	15
CE7	6	6	6	6	6
CE8	14	14	14	14	13
CE9	9	9	9	10	10
CE10	10	10	10	9	9
CE11	14	14	14	14	13
CE12	4	4	4	4	4
CE13	5	5	5	5	5
CE14	3	3	3	3	3
CE15	2	1	2	2	1

The sensitivity analysis conducted underscores the fluctuations that occur due to alterations in the weights that have been assigned. The consistency of R's practices CE15 (top-ranked Circular Economy) after varying weights given to the five experts highlights the significance of MCDM methods to provide results that will remain resilient to small changes in weights given to experts. When MOORA and EDAS MCDM methods are subject to changes in weights given to the decision matrix the rankings obtain are slightly altered.

6.0 Discussions

Rankings obtained from two unique methods corroborates the reliability of MCDM techniques. The R's practices (reduce, reuse, remanufacture, recycle) (CE_{15}), is identified as the top ranked practice. This highlights the significance of R's practices in sustainability goals. For Indian organizations aiming towards optimizing present resources to cater future requirements should adapt to R's practices. Problems aggregating due to rapid growth in industrialization in India can mitigated by underscoring the impact of this cornerstone circular economy practice in near future [19]. Prioritizing R's practices (CE_{15}) in both public and private sectors alongside applying artificial intelligence can secure long-term ambitions for achieving sustainable development.

7.0 Implications to Theory/Practice/Policy

The Indian government has taken proactive steps toward this transition, establishing the Circular Economy Cell in NITI Aayog and implementing sector-specific action plans and policies aimed at promoting resource efficiency and sustainable practices[22].India's circular economy policy include several types of mission-mode programs, that include the Swachh Bharat Mission and the E-Waste Management Rules, in along with policy frameworks like the Vehicle Scrappage Policy and Extended Producer Responsibility (EPR)[21]. The road to Viksit Bharat 2047 is green, inclusive, and circular. By embedding sustainability into flagship missions, India is positioning its youth for high-impact careers that combine innovation, environmental responsibility, and economic growth [21].

8.0 Conclusion

Upon comparing the results after calculations, it is clear that Multi-Criteria Decision-Making (MCDM) methodologies, MOORA [3], [7], [14] and EDAS [6], [8], [10], provide the same rankings for circular economy (CE) policies, fostering the practical usability of in policy making [2]. The structured approach used in decision-making process for sustainable development highlights the importance of expert's evaluation [4], [11], [13], [16]. Furthermore, the research work confirms the applicability of adopting an appropriate MCDM method that judges the expert's scores to obtain a final result that must be opted for sustainable development goals [18], [19].

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3.4.1 Limitations and Future Research Directions

This work integrates a technique to close the gap between the inspection of circular economy practices [4], [11] and the real-world adoption of Multi-Criteria Decision-Making (MCDM) procedures, such as MOORA [3], [7], [14], and EDAS [6], [8], [10]. Assessing CE policies for decision-making for sustainable development [1], [13], [17] is done and is validated by different approaches.

The limited expert evaluations and a small sample size bias the findings of the CE practices rankings. This constrains the nature of evaluation and reduces the robustness of the overall evaluation done using MCDM methodologies [2], [9].

Aiming effective environmental preservation can be secured for sustainable development by integrating MCDM methods. Using these approaches, the rapid industrialization may shape significantly applying frameworks for decision-making process that would enhance policy-making and improve robust implementation of CE practices [12], [18]. Combining digital technology with MCDM methods can further provide structured for circular economies [5], [15], [16], [19].

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